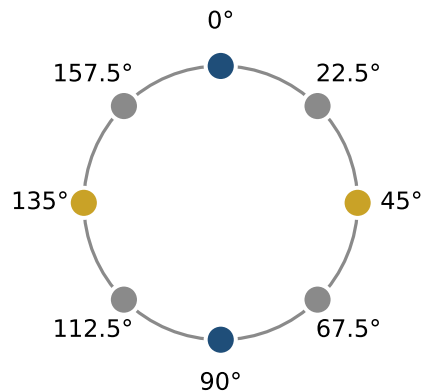


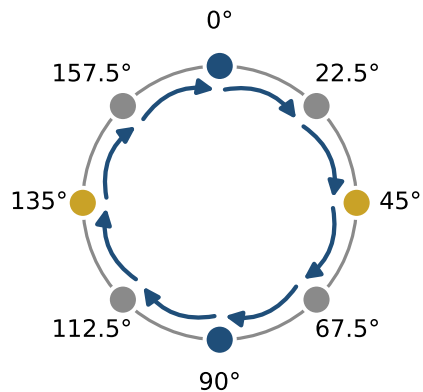
The symmetries of the orientation ring (D8: 8 rotations $\times$ 2 reflections = 16) and what the cortical anisotropy removes

(a) eight orientation classes on a ring



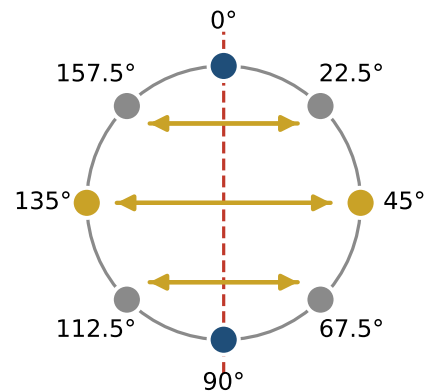
orientation is periodic in 180°, so the ring position is the doubled angle; 0° and 90° are opposite

(b) 8 rotations: shift every class



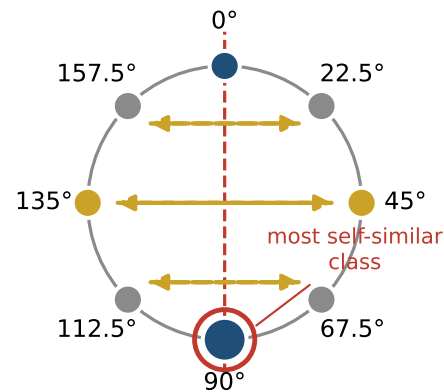
a circulant class-Gram looks the same after any shift of the labels around the ring (8, with identity)

(c) 8 reflections: mirror across an axis



the mirror through 0° and 90° swaps 45° with 135°; seven other axes give the other reflections

(d) one marked class: no rotations, one mirror left



the frame's rotation is fixed; the reflection through 90° survives and still swaps each class with its mirror